

QUERY 1;

INNER JOIN, WHERE, ORDER BY, CONCAT, UPPER

```
SELECT

c.customer_id,

UPPER(c.customer_name) AS customer_name,

c.phone,

c.address,

v.vehicle_number,

CONCAT(v.brand, ' ', v.model) AS vehicle_name,

v.vehicle_type,

it.type_name,

p.policy_id,

p.premium,

p.start_date,

p.end_date,

a.agent_name,

pay.payment_mode,

pay.amount,

cl.claim_amount,

cl.status,

g.garage_name,

r.repair_cost,

rn.renewal_date,
```

CASE

WHEN f.rating=5 THEN 'Excellent'

WHEN f.rating=4 THEN 'Good'

WHEN f.rating=3 THEN 'Average'

ELSE 'Poor'

END AS Feedback,

LENGTH(c.customer_name) AS Name_Length

FROM customer c

INNER JOIN vehicle v

ON c.customer_id=v.customer_id

INNER JOIN insurance_policy p

ON c.customer_id=p.customer_id

INNER JOIN insurance_type it

ON p.type_id=it.type_id

INNER JOIN agent a

ON p.agent_id=a.agent_id

LEFT JOIN payment pay

ON p.policy_id=pay.policy_id

LEFT JOIN claim cl

ON p.policy_id=cl.policy_id

LEFT JOIN repair r

ON cl.claim_id=r.claim_id

LEFT JOIN garage g

ON r.garage_id=g.garage_id

LEFT JOIN renewal rn

ON p.policy_id=rn.policy_id

LEFT JOIN feedback f

ON c.customer_id=f.customer_id

WHERE p.premium>10000

ORDER BY p.premium DESC;

OUTPUT:

Result Grid														
Filter Rows:														
Exports: Wrap Cell Contents:														
	customer_id	customer_name	phone	address	vehicle_number	vehicle_name	vehicle_type	type_name	policy_id	premium	start_date	end_date	agent_name	paym
3		KARTHIK	9876543212	Coimbatore	TN03EF9876	Hyundai i20	Car	Zero Depreciation	503	30000	2026-03-01	2027-03-01	Priya	Card
1		ARUN	9876543210	Chennai	TN01AB1234	Toyota Innova	Car	Comprehensive	501	25000	2026-01-01	2027-01-01	Ravi	UPI
7		SURYA	9876543216	Chennai	TN07NO3333	Kia Seltos	Car	Accident Cover	507	22000	2026-07-01	2027-07-01	Manoj	UPI
6		PRIYA	9876543215	Erode	TN06LM2222	TVS Apache	Bike	Vehicle Protection	506	18000	2026-06-01	2027-06-01	Deepa	Card

QUERY 2:

LEFT JOIN, IFNULL, CASE

SELECT

b.branch_name,

COUNT(e.employee_id) AS Total_Employees,

SUM(e.salary) AS Total_Salary,

AVG(e.salary) AS Average_Salary,

MAX(e.salary) AS Highest_Salary,

MIN(e.salary) AS Lowest_Salary,

ROUND(AVG(e.salary),2) AS Rounded_Average

FROM branch b

INNER JOIN employee e

ON b.branch_id=e.branch_id

GROUP BY b.branch_name

HAVING AVG(e.salary)>40000

ORDER BY Average_Salary DESC;

OUTPUT:

Result Grid  Filter Rows: Export:  Wrap Cell Content: 							
	branch_name	Total_Employees	Total_Salary	Average_Salary	Highest_Salary	Lowest_Salary	Rounded_Average
▶	Coimbatore South	1	70000	70000.0000	70000	70000	70000.00
	Erode Branch	1	65000	65000.0000	65000	65000	65000.00
	Chennai Branch	1	60000	60000.0000	60000	60000	60000.00
	Trichy Branch	1	50000	50000.0000	50000	50000	50000.00
	Coimbatore Branch	1	45000	45000.0000	45000	45000	45000.00
	Chennai South	1	45000	45000.0000	45000	45000	45000.00
	Chennai North	1	42000	42000.0000	42000	42000	42000.00

QUERY 3:

GROUP BY, HAVING, COUNT, AVG, SUM

SELECT

c.customer_name,

v.vehicle_number,

p.policy_id,

p.premium,

pay.payment_mode,

cl.claim_amount,

cl.status,

YEAR(cl.claim_date) AS claim_year,

MONTH(cl.claim_date) AS claim_month

FROM customer c

INNER JOIN vehicle v

ON c.customer_id = v.customer_id

INNER JOIN insurance_policy p

ON c.customer_id = p.customer_id

INNER JOIN payment pay

ON p.policy_id = pay.policy_id

INNER JOIN claim cl

ON p.policy_id = cl.policy_id

WHERE cl.claim_amount > 5000

ORDER BY cl.claim_amount DESC;

OUTPUT:

Result Grid   Filter Rows: Export:  Wrap Cell Content: 									
	customer_name	vehicle_number	policy_id	premium	payment_mode	claim_amount	status	claim_year	claim_month
▶	Karthik	TN03EF9876	503	30000	Card	20000	Approved	2026	6
	Arun	TN01AB1234	501	25000	UPI	15000	Approved	2026	5
	Priya	TN06LM2222	506	18000	Card	12000	Approved	2026	7
	Surya	TN07NO3333	507	22000	UPI	10000	Rejected	2026	8
	Vijay	TN05JK1111	505	12000	Cash	9000	Pending	2026	7
	Ajay	TN09RS5555	509	14000	Card	8000	Approved	2026	10
	Meena	TN04GH4567	504	15000	UPI	7000	Rejected	2026	6
	Keerthi	TN10TU6666	510	16000	UPI	6000	Pending	2026	11

QUERY 4:

Multiple JOIN (6–8 tables), DATE functions

SELECT

it.type_name,

COUNT(p.policy_id) AS total_policy,

SUM(p.premium) AS total_premium,

AVG(p.premium) AS average_premium,

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MAX(p.premium) AS highest_premium,

MIN(p.premium) AS lowest_premium,

ROUND(AVG(p.premium),2) AS rounded_average

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FROM insurance_type it

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INNER JOIN insurance_policy p

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ON it.type_id = p.type_id

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GROUP BY it.type_name

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HAVING AVG(p.premium) > 10000

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ORDER BY average_premium DESC;

```

OUTPUT:

Result Grid		Filter Rows:		Export:	Wrap Cell Content:		
	type_name	total_policy	total_premium	average_premium	highest_premium	lowest_premium	rounded_average
▶	Zero Depreciation	1	30000	30000.0000	30000	30000	30000.00
	Comprehensive	1	25000	25000.0000	25000	25000	25000.00
	Accident Cover	1	22000	22000.0000	22000	22000	22000.00
	Vehicle Protection	1	18000	18000.0000	18000	18000	18000.00
	Complete Cover	1	16000	16000.0000	16000	16000	16000.00
	Basic	1	15000	15000.0000	15000	15000	15000.00
	Theft Cover	1	14000	14000.0000	14000	14000	14000.00
	Premium	1	12000	12000.0000	12000	12000	12000.00

QUERY 5:

Subquery + Aggregate

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SELECT

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    c.customer_name,

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    v.vehicle_number,

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    g.garage_name,

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    r.repair_cost,

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    cl.status,  
  
    UPPER(g.garage_name) AS garage,  
  
    LENGTH(c.customer_name) AS name_length  
  
FROM customer c  
  
INNER JOIN vehicle v  
  
ON c.customer_id = v.customer_id  
  
INNER JOIN insurance_policy p  
  
ON c.customer_id = p.customer_id  
  
INNER JOIN claim cl  
  
ON p.policy_id = cl.policy_id  
  
INNER JOIN repair r  
  
ON cl.claim_id = r.claim_id  
  
INNER JOIN garage g  
  
ON r.garage_id = g.garage_id  
  
WHERE r.repair_cost >  
  
(  
  
SELECT AVG(repair_cost)  
  
FROM repair  
  
);
```

OUTPUT:

Result Grid   Filter Rows: Export:  Wrap Cell Content: 							
	customer_name	vehicle_number	garage_name	repair_cost	status	garage	name_length
▶	Arun	TN01AB1234	ABC Garage	12000	Approved	ABC GARAGE	4
	Karthik	TN03EF9876	Star Garage	18000	Approved	STAR GARAGE	7
	Priya	TN06LM2222	Royal Garage	11000	Approved	ROYAL GARAGE	5
	Surya	TN07NO3333	City Garage	9000	Rejected	CITY GARAGE	5

QUERY 6:

Correlated Subquery

SELECT

c.customer_name,

v.vehicle_number,

p.policy_id,

p.premium,

it.type_name,

a.agent_name

FROM customer c

INNER JOIN vehicle v

ON c.customer_id = v.customer_id

INNER JOIN insurance_policy p

ON c.customer_id = p.customer_id

INNER JOIN insurance_type it

ON p.type_id = it.type_id

INNER JOIN agent a

ON p.agent_id = a.agent_id

WHERE p.premium BETWEEN 10000 AND 25000

AND c.customer_name LIKE 'A%'

OR c.customer_name LIKE 'M%'

ORDER BY p.premium DESC;

OUTPUT:

Result Grid  Filter Rows: Export:  Wrap Cell Content: 						
	customer_name	vehicle_number	policy_id	premium	type_name	agent_name
▶	Arun	TN01AB1234	501	25000	Comprehensive	Ravi
	Meena	TN04GH4567	504	15000	Basic	Kumar
	Ajay	TN09RS5555	509	14000	Theft Cover	Ram

QUERY 7:

Functions (RANK(), ROW_NUMBER())

SELECT

b.branch_name,

e.employee_id,

e.name,

e.role,

e.salary,

RANK() OVER(ORDER BY e.salary DESC) AS salary_rank,

ROW_NUMBER() OVER(ORDER BY e.salary DESC) AS row_no

FROM employee e

INNER JOIN branch b

ON e.branch_id = b.branch_id

WHERE e.salary > 35000

ORDER BY e.salary DESC;

OUTPUT:

Result Grid							
		Filter Rows:		Export:	Wrap Cell Content:		
	branch_name	employee_id	name	role	salary	salary_rank	row_no
►	Coimbatore South	10	Pooja	Manager	70000	1	1
	Erode Branch	5	Riya	Manager	65000	2	2
	Chennai Branch	1	Anu	Manager	60000	3	3
	Trichy Branch	8	Divya	HR	50000	4	4
	Coimbatore Branch	3	Sita	Agent	45000	5	5
	Chennai South	7	Mohan	Agent	45000	5	6
	Chennai North	6	Ajith	Staff	42000	7	7
	Madurai Branch	2	Rahul	Staff	40000	8	8
	Madurai South	9	Sanjay	Staff	38000	9	9

QUERY 8:

LIMIT, OFFSET, DISTINCT

SELECT

c.customer_name,

f.rating,

f.comments,

CASE

WHEN f.rating = 5 THEN 'Excellent'

WHEN f.rating = 4 THEN 'Good'

WHEN f.rating = 3 THEN 'Average'

ELSE 'Poor'

END AS feedback_status

FROM customer c

INNER JOIN feedback f

ON c.customer_id = f.customer_id

ORDER BY f.rating DESC;

OUTPUT:

Result Grid				
		Filter Rows:	Export: 	
	customer_name	rating	comments	feedback_status
▶	Arun	5	Good service	Excellent
	Karthik	5	Excellent	Excellent
	Priya	5	Very good	Excellent
	Ajay	5	Excellent service	Excellent
	Divya	4	Fast response	Good
	Vijay	4	Nice support	Good
	Nisha	4	Good	Good
	Meena	3	Average	Average
	Keerthi	3	Okay	Average
	Surva	2	Need improvement	Poor

QUERY 9:

String Functions (UPPER, LOWER, LENGTH, CONCAT)

SELECT

c.customer_name,

v.vehicle_number,

p.policy_id,

p.premium,

pay.payment_mode,

pay.amount

FROM customer c

INNER JOIN vehicle v

ON c.customer_id = v.customer_id

INNER JOIN insurance_policy p

ON c.customer_id = p.customer_id

INNER JOIN payment pay

ON p.policy_id = pay.policy_id

ORDER BY p.premium DESC

LIMIT 5 OFFSET 0;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content: IA Fel
	customer_name	vehicle_number	policy_id	premium	payment_mode	amount
►	Karthik	TN03EF9876	503	30000	Card	30000
	Arun	TN01AB1234	501	25000	UPI	25000
	Surya	TN07NO3333	507	22000	UPI	22000
	Priya	TN06LM2222	506	18000	Card	18000
	Keerthi	TN10TU6666	510	16000	UPI	16000

QUERY 10:

CASE, IFNULL, ORDER BY and aggregate functions

SELECT

c.customer_name,

CONCAT(v.brand,' ',v.model) AS vehicle_name,

p.policy_id,

p.premium,

pay.payment_mode,

IFNULL(cl.claim_amount,0) AS claim_amount,

IFNULL(r.repair_cost,0) AS repair_cost,

rn.renewal_date,

UPPER(a.agent_name) AS agent_name

FROM customer c

INNER JOIN vehicle v

ON c.customer_id = v.customer_id

INNER JOIN insurance_policy p

ON c.customer_id = p.customer_id

INNER JOIN agent a

ON p.agent_id = a.agent_id

LEFT JOIN payment pay

ON p.policy_id = pay.policy_id

LEFT JOIN claim cl

ON p.policy_id = cl.policy_id

LEFT JOIN repair r

ON cl.claim_id = r.claim_id

LEFT JOIN renewal rn

ON p.policy_id = rn.policy_id

ORDER BY p.premium DESC;

OUTPUT:

Result Grid Filter Rows: Export: Wrap Cell Content:									
	customer_name	vehicle_name	policy_id	premium	payment_mode	claim_amount	repair_cost	renewal_date	agent_name
▶	Karthik	Hyundai i20	503	30000	Card	20000	18000	2027-03-01	PRIYA
	Arun	Toyota Innova	501	25000	UPI	15000	12000	2027-01-01	RAVI
	Surya	Kia Seltos	507	22000	UPI	10000	9000	2027-07-01	MANOJ
	Priya	TVS Apache	506	18000	Card	12000	11000	2027-06-01	DEEPA
	Keerthi	Bajaj Pulsar	510	16000	UPI	6000	5000	2027-10-01	ANITHA
	Meena	Honda Shine	504	15000	UPI	7000	6000	2027-04-01	KUMAR
	Ajay	Maruti Swift	509	14000	Card	8000	7000	2027-09-01	RAM
	Vijay	Ford EcoSport	505	12000	Cash	9000	8000	2027-05-01	LAKSHMI
	Nisha	Royal Classic	508	9000	Cash	4000	3000	2027-08-01	HARI
	Divya	Yamaha B15	502	8000	Cash	5000	4000	2027-12-01	SIDESH